Jungpyo Lee, Ph.D.

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Academic Appointments

Aug. 2024 – present **Postdoctoral Scholar**, University of California, Berkeley
Department of Mechanical Engineering

Education

2019 – 2024 **Ph.D. in Mechanical Engineering**, University of California, Berkeley, USA
2015 – 2017 **M.Sc. in Mechanical Engineering**, Yonsei University, Seoul, South Korea
2009 – 2015 **B.Sc. in Mechanical Engineering**, Yonsei University, Seoul, South Korea

Research Experience

Aug. 2019 – present **Embodied Dexterity Group**, University of California, Berkeley, USA
Doctoral Researcher → Postdoctoral Scholar; Advisor: Prof. Hannah Stuart
Research: Tactile sensing and robotic grasping; wearable assistive devices and human–robot collaboration.

Apr. 2015 – May 2019 **Center for BioMicrosystems**, Korea Institute of Science and Technology, Seoul, South Korea
Research Trainee (2015) → Commissioned Researcher (2018); Advisor: Dr. Il-Joo Cho
Research: Brain–machine interfaces, biosensors, and neuromodulation.

Mar. 2015 – Aug. 2017 **Nano Electro-Mechanical Device Lab**, Yonsei University, Seoul, South Korea
Master's Research; Advisor: Prof. Seong Chan Jun
Research: Nanomaterials and neural connectivity.

Mar. 2014 – Feb. 2016 **Astrodynamics and Control Lab**, Yonsei University, Seoul, South Korea
Engineer, Structural and Thermal Control Systems; Advisor: Prof. Sang-Young Park
Research: CANYVAL-X mission.

Honors & Awards

- 2025: Best Presentation, 3rd Stanford x Berkeley Korean Researcher Academic Conference.
- 2024: Fung Institute Capstone Technical Leadership Award.
- 2024: Best Presentation, 2nd Stanford x Berkeley Korean Researcher Academic Conference.
- 2018: Korean Government Scholarship for Graduate Study Overseas, 2019–2021.
- 2017: Best Paper Award, Oral Session, 19th Korean MEMS Conference.
- 2017: Best Paper Award, School of Mechanical Engineering, Yonsei University.

Publications

ORCID: 0000-0002-2949-7773 Google Scholar

* Equal contribution. † Corresponding author.

Peer-Reviewed Journal Articles

J8 **Lee**, **J.**, McPherson, A. I. W., Huang, H., Yu, L., Gloumakov, Y., & Stuart, H. S. (2025). Expanding Functional Workspace for People With C5-C7 Spinal Cord Injury With Supernumerary Dorsal Grasping. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, *33*, 22–33. **Featured article for February 2025: link**. doi:10.1109/TNSRE.2024.3514135. (Preprint)

J7 **Lee***, **J.**, Lee*, S. D., Huh, T. M., & Stuart, H. S. (2024). Haptic search with the Smart Suction Cup on adversarial objects. *IEEE Transactions on Robotics*, 226–239. **Spotlighted article in Science Robotics' "Editor's Choice": link**. doi:10.1109/TRO.2023.3331063. (Preprint)

J6 Jeong, U.-J., **Lee**, **J.**, Chou, N., Kim, K., Shin, H., Yu, H.-Y., & Cho, I.-j. (2021). A minimally invasive flexible electrode array for simultaneous recording of ECoG signals from Multiple brain regions. *Lab on a chip*, 2383–2397. doi:10.1039/D1LC00117E

J5 Oh, S.-J., Lee, J. M., Kim, H.-B., **Lee**, **J.**, Han, S., Bae, J. Y., Hong, G.-S., Koh, W., Kwon, J., Hwang, E.-S. et al. (2019). Ultrasonic neuromodulation via astrocytic TRPA1. *Current Biology*, *29*(20), 3386–3401. doi:10.1016/j.cub.2019.08.021

J4 **Lee***, **J.**, Ko*, K., Shin, H. [Hyogeun], Oh, S.-J., Lee, C. J., Chou, N., Choi, N., Tack Oh, M., Chul Lee, B., Chan Jun, S. et al. (2019). A MEMS ultrasound stimulation system for modulation of neural circuits with high spatial resolution in vitro. *Microsystems & Nanoengineering*, *5*(1), 28. doi:10.1038/s41378-019-0070-5

J3 Kundu, A., **Lee**, **J.**, Park, B., Ray, C., Sankar, K. V., Kim, W. S., Lee, S. H., Cho, I.-J., & Jun, S. C. (2018). Facile approach to synthesize highly fluorescent multicolor emissive carbon dots via surface functionalization for cellular imaging. *Journal of colloid and interface science*, *513*, 505–514. **Cover Page**. doi:10.1016/j.jcis.2017.10.095

J2 Ray, C., Lee, S. C., Sankar, K. V., Jin, B., **Lee**, **J.**, Park, J. H., & Jun, S. C. (2017). Amorphous phosphorus-incorporated cobalt molybdenum sulfide on carbon cloth: An efficient and stable electrocatalyst for enhanced overall water splitting over entire ph values. *ACS applied materials & interfaces*, *9*(43), 37739–37749. doi:10.1021/acsami.7b11192

J1 Some, S., Sohn, J. S., Kim, J., Lee, S.-H., Lee, S. C., **Lee**, **J.**, Shackery, I., Kim, S. K., Kim, S. H., Choi, N. et al. (2016). Graphene-iodine nanocomposites: Highly potent bacterial inhibitors that are bio-compatible with human cells. *Scientific reports*, *6*(1), 20015. doi:10.1038/srep20015

Peer-Reviewed Conference Proceedings

C6 Aderibigbe, J., Li, M., **Lee**, **J.**, & Stuart, H. S. (2025). Milli-scale acoustac sensing using soft helmholtz resonators. In *2025 IEEE International Conference on Robotics and Automation (ICRA)* (pp. 16585–16590). IEEE. doi:10.1109/ICRA55743.2025.11128844

C5 Lee, S., **Lee**, **J.**, & Stuart, H. S. (2024, October). Haptic contour following with the smart suction cup. In *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 5232–5237). IEEE. doi:10.1109/IROS58592.2024.10802260

C4 **Lee**, **J.**, Sun, Z., Dong, Z., Chen, F., & Stuart, H. S. (2024, August). Regrasping on printed circuit boards with the smart suction cup. In *2024 IEEE International Conference on Robotics and Automation (ICRA)* (pp. 6477–6483). IEEE. doi:10.1109/ICRA57147.2024.10610153

C3 **Lee**, **J.**, Yu, L., Derbier, L., & Stuart, H. S. (2021, October). Assistive supernumerary grasping with the back of the hand. In *2021 IEEE International Conference on Robotics and Automation (ICRA)* (pp. 6154–6160). IEEE. doi:10.1109/ICRA48506.2021.9560949

C2 Park, J. P., Park, S. Y., Song, Y. B., Kim, G. N., Lee, K., Oh, H. J., Yim, J. C., Lee, E. [Eunji], Hwang, S. H., Kim, S. [Sungwoo] et al. (2016, May). Cubesat development for canyval-x mission. In *14th international conference on space operations, spaceops 2016* (AIAA–2016). AIAA. doi:10.2514/6.2016-2493

C1 Chae, U., Shin, H. [Hyoguen], Lee, H. J., **Lee**, **J.**, Choi, N., Lee, Y. J., Lee, S. H., Woo, J., Cho, Y., Yoon, E.-S. et al. (2016, February). A new mems neural probe system integrated with push-pull microfluidic channels and biosensors for real-time monitoring of neurochemicals. In *2016 IEEE 29th International Conference on Micro Electro Mechanical Systems (MEMS)* (pp. 329–332). IEEE. doi:10.1109/MEMSYS.2016.7421627

Accepted Papers

Jeong, G., **Lee**†, **J.**, & Stuart, H. S. Enhanced haptic search with a multi-chamber suction cup using reverse flow. In *2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. **Accepted**. IEEE.

Manuscripts Under Review

Lee, **J.**, McPherson, A. I. W., Akhavan, A., & Stuart, H. S. A Multi-Dimensional Evaluation of Assistive Wearable Grasping Devices in unstructured Daily Environments. **Under Review**.

Lee, **J.**, & Stuart, H. S. Co-designing suction cup structure and haptic control policy for adaptive robotic grasping. **Under Review**.

Lee*, **J.**, McPherson*, A. I. W., Ma, D., & Stuart, H. S. Stow-able Passive Supernumerary Dorsal Grasping. **Under Review**.

Manuscripts in Preparation

Davis*, B., **Lee***, **J.**, & Stuart, H. S. Hybrid Direct-Drive and Underactuated Linkage-Driven Dexterous Hand for Efficient Grasping and Manipulation.

Lee, **J.**, Chen, S.-J., Sudhir, K., Stessman, L., & Stuart, H. S. AI-Driven Personalized Assistive Wearable Grasper for Human Variability.

Preprints and Technical Reports

1 **Lee**, **J.**, Lee, S. D., Huh, T. M., & Stuart, H. S. (2024, January). Initial analysis of data-driven haptic search for the smart suction cup. ArXiv. doi:10.48550/arXiv.2401.06354

Non-Refereed Conference Proceedings

2 Chase, U., Shin, H., **Lee**, **J.**, Choi, N., Lee, Y. J., Lee, S. H., Yoon, E.-S., Yu, H.-Y., & Cho, I.-J. (2016, December). A new mems neural probe system for real-time monitoring of neurotransmitters and neural signals. In *2016 Korean Mechanical Engineering Conference Fall Meeting* (pp. 3028–3028).

1 Chase, U., Shin, H., Lee, H. J., **Lee**, **J.**, Choi, N., Lee, Y. J., Lee, S. H., Yoon, E.-S., Yu, H.-Y., & Cho, I.-J. (2015, November). A new mems neural probe system integrated with push-pull microfluidic channels. In *70th Anniversary Korean Mechanical Engineering Conference* (pp. 2232–2232).

Theses

2 **Lee**, **J.** (2024). *Enhancing grasping in robotic and human-robot systems by leveraging intrinsic functionality*. Ph.D. Dissertation. Retrieved from https://www.proquest.com/docview/3110962183?pq-origsite=gscholar&fromopenview=true&sourcetype=Dissertations%20&%20Theses

1 **Lee**, **J.** (2017). *Enhancement of functional connectivity in three-dimensional neuronal culture*. Master's thesis, Yonsei University. Retrieved from https://library.yonsei.ac.kr/search/detail/CATTOT000001862381

Book Chapters

1 Park, J.-P., Park, S.-Y., Song, Y. B., Kim, G. N., Lee, K., Oh, H. J., Yim, J.-C., Lee, E. [EunJi], Hwang, S.-H., Kim, S. [SungWoo] et al. (2017). *Canyval-x mission development using cubesats*. doi:10.1007/978-3-319-51941-8_30

Patents and Patent Applications

1 Jun, S. C., Park, B., & **Lee**, **J.** (2017, October). Tonometer using fret. App. No. 1020150092327 Korean Patent.

Research Presentations

Invited Talks

2 From Brains to Robots: My Journey Through Research and Life. (2025, December), In *2025 End of the Year Seminar at Korean-American Scientists and Engineers Association (KSEA) Berkeley Chapter*, Berkeley, CA.

1 Grappling with the Environment: Design of Forceful Mobile Robots for Uncertain Planetary Surfaces. (2025, April), In *The HARDER Workshop*. Robosoft 2025, Lausanne, Switzerland.

Seminars and Research Talks

8 From Additive Complexity to Intrinsic Intelligence: Designing Reliable Robotic Manipulation for the Real World. (2026, March), In *North Carolina State University*, Raleigh, NC.

7 Co-designing Multi-Chambered Suction Cups and Haptic Search Controller for Adaptive Robot Grasping. (2026, January), In *Gordon Research Seminar*, Ventura, CA.

6 Failure-Resilient Grasping Through Inherent Functionality: Autonomous and Assistive Perspectives. (2025, December), In *Research Seminar at Korean-American Roboticists Association (KARA)*, Online.

5 Grasping Together: Human-Robot Collaboration for Functional Recovery. (2025, November), In *The 3rd Berkeley x Stanford Academic Conference and Networking (organized by Korean Graduate Student Association, powered by Samsung)*, San Jose, CA.

4 Enhancing grasping in robotic and human-robot systems by leveraging intrinsic functionality. (2025a, April), In *Yonsei University*, Seoul, Korea.

3 Enhancing grasping in robotic and human-robot systems by leveraging intrinsic functionality. (2025b, April), In *Sogang University*, Seoul, Korea.

2 Enhancing grasping in robotic and human-robot systems by leveraging intrinsic functionality. (2025c, April), In *Sejong University*, Seoul, Korea.

1 Enhancing grasping in robotic systems by leveraging intrinsic functionality. (2024, November), In *The 2nd Berkeley x Stanford Academic Conference and Networking (organized by Korean Graduate Student Association, powered by Samsung)*, San Jose, CA.

Workshop and Conference Abstracts

9 **Lee**, **J.**, McPherson, A. I. W., Akhavan, A., & Stuart, H. S. (2026, September 27). Beyond Structured Testing: Evaluating Assistive Wearable Grasping Devices in Unstructured Daily Environments. In *IROS 2026 Workshop: Out of the Lab: Personalization and Long-Term Use in Rehabilitation and Assistive Robotics*.

8 **Lee**, **J.**, Lee, S., & Stuart, H. S. (2024, May). Investigating the Impact of Chamber Variations on Smart Suction Cup Deformation and Grasping Efficiency. In *RoboSoft 2024: Late Breaking Results*.

7 **Lee**, **J.**, Lee, S., & Stuart, H. S. (2023a, October). Model-based Tactile Regrasping with the Smart Suction Cup. In *IROS 2023 Workshop: RoboTac: Visuo-Tactile Perception, Learning, Control for Manipulation and HRI*.

6 **Lee**, **J.**, Lee, S., & Stuart, H. S. (2023b, October). Model-based Tactile Regrasping with the Smart Suction Cup. In *IROS 2023: Late Breaking Results*.

5 **Lee**, **J.**, McPherson, A., Huang, H., Gloumakov, Y., & Stuart, H. S. (2023a, October). The Dorsal Grasper 2.0: Supernumerary robotic grasping for people with C5/C6 Spinal Cord Injury. In *IROS 2023 Workshop: Assistive Robotics for Citizens*.

4 **Lee**, **J.**, McPherson, A., Huang, H., Gloumakov, Y., & Stuart, H. S. (2023b, October). The Dorsal Grasper 2.0: Supernumerary robotic grasping for people with C5/C6 Spinal Cord Injury. In *IROS 2023: Late Breaking Results*.

3 Chase, U., Shin, H., **Lee**, **J.**, Choi, N., Lee, Y. J., Lee, S. H., Yoon, E.-S., Yu, H.-Y., & Cho, I.-J. (2017, April). A new MEMS neural probe system for real-time monitoring of neurotransmitters and neural signals. In *KMEMS 2017*.

2 **Lee**, **J.**, Ko, K., Lee, J.-H., Oh, S.-J., Shin, H., Lee, B. C., Lee, C. J., Kim, T. s., Choi, N., Jun, S. C., & Cho, I.-J. (2017, April). MEMS Ultrasonic Transducer Array for Regulating Neural Circuits through Stimulation of Neurons. In *KMEMS 2017*.

1 **Lee**, **J.**, Ko, K., Lee, J.-H., Oh, S.-J., Shin, H., Lee, C. J., Kim, T. s., Choi, N., Jun, S. C., & Cho, I.-J. (2016, September). Micromachined Piezoelectric Transducer Array for Localized ultrasound Stimulation of Brain with High Spatial Resolution. In *19th Korean Society for Brain and Neural Science Annual Meeting*.

Teaching Experience

University of California, Berkeley

Graduate Student Instructor (Teaching Assistant)

- **Mechatronics Design (ME 102B)**
  Fall 2021
  Laboratory instruction: mechatronics, ESP32, Arduino, mechanical design, and project management.
  Course instructor: Prof. Hannah Stuart.
- **Advanced Engineering Graphical Communication (E 128 / ME 292C)**
  Fall 2020
  Laboratory instruction: Creo, 3ds Max, and project management.
  Course instructor: Prof. Dennis K. Lieu.

- **Visualization for Design (E 25)**
  Spring and Fall 2020
  Laboratory instruction: 2D engineering drawing, AutoCAD, and GD&T.
  Course instructor: Prof. Dennis K. Lieu.

Research Mentoring

Current Graduate Students

Jadesola Aderibigbe (UC Berkeley), Gia Jeong (UC Berkeley), Isabel Morales (UC Berkeley), Sun Zheng (CUHK), Ben Davis (UC Berkeley)

Current Undergraduate Students

Jeongbin Sohn (GIST), Alex Chen (UC Berkeley), Ninad Atale (UC Berkeley), Emiliano Sanchez (UC Berkeley), Meaghan Pearson (UC Berkeley)

Past Berkeley Master of Engineering (MEng)

Shou-Jen Chen, Lilyane Stessman, Krishnaa Sudhir (2025-2026)
- AI-Driven Design Optimization for Personalized Assistive Robots

Andrew Darmitzel, Michael Rubin, Li Li, Candice Xia (2025-2026)
- Open Hardware for Robotic Trash Sorting – Next Generation Smart Suction Cup

Jessica Boetticher, Adam Duong, Benjamin Margolis, Panos Pardalidis (2023-2024)
- Next generation assistive wearable for everyday grasping with Spinal Cord Injury
- Fung Institute Capstone Technical Leadership Award: link

Past Undergraduate Students

- Alahe Akhavan, Graduate Student at Johns Hopkins University
- Licheng Yu, Product & Process Development Engineer at Foxconn
- Haoxiang (Lucas) Hung, Recycling Program Manager at Triple L Global
- Derrick Ma, PhD student at Caltech

Professional Service

Editorial Service

- **Associate Editor**, IEEE ICRA 2027 — Manipulation and Grasping 2.

Peer Review

- **Journals:** IEEE Transactions on Robotics; IEEE/ASME Transactions on Mechatronics; IEEE Robotics and Automation Letters; Journal of Computational Design and Engineering; Frontiers in Aging Neuroscience; Frontiers in Robotics and AI; Frontiers in Medicine.
- **Conferences:** IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).

Professional Societies

- Institute of Electrical and Electronics Engineers (IEEE) since 2020
  → Active member of the IEEE Robotics and Automation Society (RAS)
- Korean-American Scientists and Engineers Association (KSEA) since 2024

Community Outreach

- 2026: Demonstrated Suction UMI and the Berkeley QUAD Hand for visitors at "Robotics Beyond the Lab," UC Berkeley College of Engineering's Homecoming Weekend, October 10. [Event highlights]
- 2026: Led a lab tour and delivered a career talk for high school students from Siheung, Korea, August 06.
- 2026: Guided undergraduate students at Korea Nazarene University through a robotics lab tour and discussed research for wearable assistive devices, February 03.
- 2026: Mentored high school students of the Lee Do Leader Program (Sejong, Korea) in robotics, leading campus and lab tours during their U.S. visit and continuing to provide academic and career guidance beyond the visit, January.
- 2025: Robotics demonstration at Rosa Parks Elementary School (Berkeley, CA) during the Robotics & Engineering Demo Day, November 13.
- 2025: Guided undergraduate students at Seoul National University (SNU) through a robotics lab tour and shared career advice during a panel session as part of the SNU at Silicon Valley program, July 29.
- 2025: Led a hands-on session with young DaVinci scholars as part of the DaVinci Camp Summer Institute, letting students try the Dorsal Grasper system, June 18.
- 2025: Guest presentation at EnableTech, a student organization that designs and builds assistive technology for people with disabilities, March 03.

Media Coverage

- 2025: Interview, Hong Kong Centre for Logistics Robotics (HKCLR), "A robotics journey inspired by brain research."
- 2025: Dorsal Grasper research featured in Berkeley Engineering, including Berkeley's Impacts of Research page and an accompanying video.
- 2025: Dorsal Grasper research featured in R&D World Magazine.